

1. PROBLEM STATEMENT

Soil health is critical to the growth of healthy crops, and knowing what nutrients are lacking is necessary to correct for deficiencies. Digital Instrument for Remote Terrain Testing (D.I.R.T.T.) is an autonomous soil testing robot that allows users to identify critical soil health markers, thereby allowing them to properly address any issues.

1.1. Need Statement

Approximately 25,000 people worldwide, including more than 10,000 children, die each day from hunger and related causes [1]. In the U.S., farmers cannot meet domestic demands; in 2025, agricultural imports exceeded agricultural exports by \$41 billion [2]. Meanwhile, farmers reported at least one soil or water resource concern on 49% of surveyed soybean, oat, cotton, and wheat fields [3]. Agricultural soil testing traditionally requires numerous individual soil cores to produce a representative composite sample. Depending on the sampling methodology, extension services generally recommend collecting approximately 10–20 cores from each management area or zone [4]. Improper sampling location, depth, or technique can produce unrepresentative samples and consequently inaccurate soil-test recommendations [5]. Current sampling methods, which are labor-intensive and error-prone, show a need for an autonomous soil-sampling system.

1.2. Objective Statement

The objective of this project is to design and prototype a device that aids in the detection of nutrient, pH, and moisture imbalances in soil, reducing the time and labor required by manual sampling. D.I.R.T.T. autonomously navigates in a grid-like pattern to retrieve soil samples, then analyzes and stores the results for the user to access.

1.3. Background and Related Work

Soil health is normally evaluated by measuring nutrient levels, such as NPK and pH. Nutrient sensing relies on ion-selective electrodes (ISEs), while pH is measured using a similar ion-selective electrode immersed in a soil solution. Autonomous systems have been developed to address this: Rogo's SmartCore follows a pre-programmed grid, covering about 500 acres in five hours while reducing sampling variability by at least 10% [6], [7]. Agrobot Lala uses satellite imagery and AI to select sampling zones, then autonomously extracts and analyzes each sample for nitrates on-site in about 30 minutes [8].

D.I.R.T.T. shares some values with both systems: autonomous, GPS-guided navigation to sampling locations, but differs in how it divides the work and how it is powered. Rogo demonstrates fast, reliable autonomous sampling, but all analysis is done in an off-site lab, and it runs on diesel [6], while Lala demonstrates in-field analysis but at a slower pace [8]. This project design combines navigation, sampling, and on-board analysis in a single solar-powered unit, positioning it as a lower-cost, fuel-independent alternative that avoids emissions, refueling, and operating costs of larger vehicles. Existing designs, such as ISE-based nitrate probes with protective membranes for field insertion [9], [10], and multi-ion electrode arrays with defined response times and service life [11], establish practical approaches that may be helpful for this design.

References

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